Personal Statement

National University of Singapore
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On a Dobot CR5, I learned a controller can appear faster yet become less reliable. Non-blocking execution at higher command frequency let each policy inference observe an intermediate arm state before the preceding action chunk finished. The resulting corrections drove the robot back and forth. I therefore used blocking, short-horizon ServoP action chunks, accepting lower apparent frequency to preserve state-action consistency and trajectory quality. This sharpened my guiding question: how can learning, control, timing, sensing, and safety form one reliable robot system?

My preparation combines complementary disciplines by design. I completed a B.Eng. in Mechanical Design, Manufacturing and Automation at Harbin Institute of Technology, Weihai, in June 2025. That degree and robotics projects grounded me in kinematics, robot modeling, and physical constraints. I am now pursuing a second bachelor's degree in Computer Science and Technology at Harbin Institute of Technology, Shenzhen, expected in June 2027. This is not a departure from mechanical engineering, but a deliberate expansion from modeling machines to building the algorithms, data pipelines, and interfaces through which they perceive, learn, and act. Together, these perspectives let me examine embodied systems from physical and computational sides.

That combination became concrete in a CR5 data-collection and inference loop using an Orbbec Astra2 RGB-D camera, an electric gripper, and ROS2. I made RGB timestamps the HDF5 master clock, aligning robot states, target actions, and gripper feedback through binary search and linear interpolation. Using Fast DDS shared memory and RGB-only capture, I reduced recorded frame drops from 27.15% to the 2.57%-4.74% range. Rather than equate throughput with quality, I paired this transport improvement with schema, motion, workspace, and slow-playback checks before using or replaying an episode. I then converted synchronized episodes to LeRobot v2.0 and adapted OpenPI for CR5, using consistent seven-dimensional state and action semantics across collection, training configuration, and WebSocket-served deployment. The client translated policy chunks into ServoP arm targets and Modbus gripper commands. This continuity mattered because a dataset field is not merely a storage choice: it defines what the policy learns to predict and what the hardware later interprets. The project therefore made reliability a chain of aligned decisions across clocks, representations, validation, and execution timing.

A complementary research thread examines action representation. As a Research Assistant at iLearn, I am second author of Hermite Action Tokenizer for VLA, an unsubmitted working paper, and I am responsible for its discrete autoregressive tokenization branch. Working on this branch made representation a control question: what trajectory structure should tokenization preserve, and how should gripper decisions be encoded separately? The method separates seven-dimensional action chunks into six-dimensional end-effector trajectories and gripper states, with shared breakpoints removing endpoint redundancy and quantization seams. In an offline codec benchmark averaging four LIBERO subsets with 8-by-7 action chunks, Hermite's Endpoint MSE was approximately 99.7% lower than FAST and 82.6% lower than BEAST. I therefore interpret the comparison narrowly: it provides evidence about offline codec fidelity under that exact protocol, not about online rollout performance or physical-robot error. The research showed that action encoding constrains what a model can express; CR5 deployment showed that execution schedules determine which state the model observes and what its command controls. Reliable embodied intelligence therefore depends on both representation semantics and execution semantics, joined through synchronized data.

During the Xbotics Embodied AI Community Internship, I migrated Isaac Lab's ANYmal-C navigation task into a MotrixLab NumPy environment. My judgment was to treat migration as interface preservation, not transcription. I refactored reset and step flows, command sampling, observations, rewards, and termination, corrected MuJoCo heightfield frames, added spawn and goal constraints, and used reward and termination regression checks. Earlier, in the HIT Weihai HERO RoboMaster Team's Vision Group, I helped modularize image acquisition, vision processing, and communication synchronization as ROS2 nodes and worked on host-MCU timing. These experiences reinforced that coordinate, interface, and timing assumptions must be explicit for integration.

My immediate goal is an R&D engineering role in robotics or embodied AI, connecting learning methods to sensing, control, and deployment. I want to strengthen my ability to design and evaluate real-hardware systems, where representation, timing, and physical constraints must be reconciled. After gaining stronger real-system experience, I may pursue doctoral study; it remains an option rather than a predetermined destination. I therefore seek graduate training whose curriculum and applied opportunities can close gaps between my integration experience and the robot systems I want to build. My Hermite work examines how action representations preserve trajectory structure. In the separate CR5 project, I learned that a policy command is meaningful on hardware only when synchronized visual observations, the robot's current state, and execution timing remain consistent. I can evaluate representation fidelity and trace deployment failures, but I need deeper command of how learning, dynamics, perception, and physical validation interact. NUS's MSc in Robotics offers that integrated training.

If the current offerings remain available in my intake, ME5409 Robot Dynamics and Control would strengthen my analysis of action timing and controlled motion; ME5411 Robot Vision and AI would deepen how visual observations support robot decisions; and ME5418 Machine Learning in Robotics would connect structured actions to learned policies. Subject to availability and approval, either the optional eight-unit ME5400 Robotics Project or four-unit ME5400A Robotics Project 1 could provide one supervised setting for two deliberately separate studies: representation fidelity evaluated under a specified offline codec protocol, and state-action consistency examined in distinct hardware trials. I would contribute experience with synchronized robot data, tokenizer evaluation, and short-horizon ServoP execution. This preparation would advance my goal of becoming a robotics and embodied-AI R&D engineer who builds learned systems in which visual evidence, action semantics, and control execution remain coherent.